THE UNIVERSITY OF QUEENSLAND
AUSTRALIA

This exam paper must not be removed from the venue

Venue _____
Seat Number _____
Student Number |__|__|__|__|__|__|__|__|
Family Name _____
First Name _____

# School of Mathematics & Physics

## Semester Two Examinations, 2024

## PHYS4055 Atomic Physics and Quantum Optics

*This paper is for St Lucia Campus students.*

**Examination Duration:** 120 minutes

**Planning Time:** 10 minutes

**Exam Conditions:**

•Any written or printed material is permitted; material may also be annotated.
•Casio FX82 series or UQ approved and labelled calculator only
•During Planning Time - Students are encouraged to review and plan responses to the exam questions

**Materials Permitted in the Exam Venue:**
***(No electronic aids are permitted e.g. laptops, phones)***

Any additional written or printed material is permitted; material may also be annotated.

**Materials to be supplied to Students:**
***Additional exam materials (e.g. answer booklets, rough paper) will be provided upon request.***

1 x 14-Page Answer Booklet

**Instructions to Students:**
***If you believe there is missing or incorrect information impacting your ability to answer any question, please state this when writing your answer.***

There are FOUR questions. Answer any THREE.

**For Examiner Use Only**

| Question | Mark |
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Total _____

**Question 1**

(10 marks)

In this question we will consider the splitting of energy levels due to the spin-orbit interaction. We will focus on the term ${}^3D$.

(a) (1 mark) What are the possible values of the total angular momentum $J$ for this term?

(b) (3 marks) The Hamiltonian for the spin-orbit interaction may be expressed as the scalar product of the spin and orbital angular momenta, $H = A\mathbf{L}\cdot\mathbf{S}$, where $A$ is a constant taken to be positive.

Show that the energy shift is given by

$$\Delta E = \frac{A}{2}\big(J(J+1) - L(L+1) - S(S+1)\big)\ , \tag{1}$$

showing all working.

(c) (4 marks) Hence, find the energy shift for all ${}^3D_J$ and sketch the original (unsplit) and split terms on an energy-level diagram. Comment on the energy difference between adjacent levels.

(d) (2 marks) Does the total number of states (including magnetic sublevels) change due to spin-orbit interaction? Show your working.

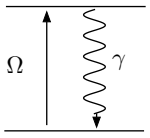
**Question 2**

(10 marks)

A Zeeman-split magnetic hyperfine transition is driven by a transverse AC magnetic field. The interaction picture Hamiltonian is

$$H = \hbar\frac{\Delta(B)}{2}\sigma_z + \Omega\sigma_x \ , \tag{2}$$

where $\sigma_z = |e\rangle\langle e| - |g\rangle\langle g|$ and $\sigma_x = |e\rangle\langle g| + |g\rangle\langle e|$, $\Delta(B)$ is the Zeeman splitting as a function of the external DC magnetic field and $\Omega$ is the Rabi frequency of the transverse driving field. The spontaneous decay rate on the transition is $\gamma$. The steady state average of $\langle\sigma_z\rangle$ is given by




$$\langle\sigma_z\rangle_{ss} = -\frac{\gamma^2 + 4\Delta(B)^2}{\gamma^2 + 4\Delta(B)^2 + 8\Omega^2} \ . \tag{3}$$

We assume that $\Delta(B) = \kappa B$.

(a) (4 marks) Calculate the steady state probability to find the atom in the ground state.

(b) (3 marks) The absorption coefficient for AC field is proportional to the steady-state probability to find the system in the ground state, $P_g$. Describe how this system could be used as a sensor for the static magnetic field at the atom.

(c) (3 marks) Show that to operate as a linear sensor we should choose a fixed external DC bias field given by

$$B_0 = \pm\frac{\sqrt{8\Omega^2 + \gamma^2}}{\sqrt{3}} \ .$$

**Question 3**

(10 marks)

A beam-splitter is an optical device with two input modes and two output modes. Consider a 50:50 beam splitter with input modes 1 and 2 and output modes 3 and 4. This beam-splitter combines the modes as $\hat{a}_3 = (\hat{a}_1 + \hat{a}_2)/\sqrt{2}$ and $\hat{a}_4 = (\hat{a}_1 - \hat{a}_2)/\sqrt{2}$ .

(a) (3 marks) What output state is expected when the two inputs each have a single photon in them $|1\rangle_1|1\rangle_2$?

(b) (2 marks) Show that for any input this beam-splitter conserves energy.

Consider the normalised state $|\psi\rangle = \alpha|n\rangle + \beta|n+1\rangle$ where $|n\rangle$ is the $n$-photon state and $|\alpha|^2 + |\beta|^2 = 1$.

(c) (3 marks) Show that this state exhibits sub-Poissonian statistics, i.e. $\mathrm{Var}(n) < \langle n \rangle$.

(d) (2 marks) Briefly describe the significance of sub-Poissonian statistics in light.

**Question 4**

(10 marks)

Consider a single optical cavity mode interacting with an external field through a cavity mirror and also experiencing internal loss. In a frame rotating at the optical frequency the quantum Langevin equation describing this cavity mode is: $\dot{\hat{a}} = -\gamma/2\ \hat{a} + \sqrt{\gamma_1}\hat{A}_{in,1} + \sqrt{\gamma_2}\hat{A}_{in,2}$, where $\gamma = \gamma_1 + \gamma_2$, and $\gamma_1$ represents the coupling through the mirror and $\gamma_2$ represents the coupling to internal loss.

(a) (2 marks) Find the solution to the above Langevin equation for the cavity mode in the Fourier domain, $\tilde{a}(\omega)$.

(b) (2 marks) Suppose the input mode $\hat{A}_{in,1}$ represents a laser beam in an unspecified state impinging on the cavity mirror. Using the solution from part (a) and the input/output relation, $\hat{A}_{out,1} = \sqrt{\gamma_1}\hat{a} - \hat{A}_{in,1}$, find the Fourier domain output mode operator for the light reflected from the cavity mirror, $\tilde{A}_{out,1}(\omega)$.

(c) (2 marks) What happens to the laser light at low Fourier frequencies ($\omega = 0$) if $\gamma_1 = \gamma_2$? What does this mean for the intensity of the reflected beam?

Suppose we wish to encode qubit information on a single photon using a dual-rail encoding.

(d) (2 marks) Describe two different physical degrees of freedom that could be used for the encoding and situations where one may be preferred over another.

(e) (2 marks) For the physical encodings discussed in part (c), describe how a Hadamard gate could be implemented in the lab for each of your encodings.

**END OF EXAMINATION**